

HYESUN YOU

Science Education | Department of Teaching and Learning | University of Iowa
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■ EDUCATION

University of Texas at Austin

Ph.D. in Science Education, 2016

M.Ed. in Quantitative Methods (Psychometrics Emphasis), 2021

Yonsei University, Seoul, South Korea

M.S. in Science Education, 2008

B.S. in Chemistry; Biology (Minor), 2005

Harvard University Online, Professional Certificate in Data Science, 2021

Johnson County Community College, Data Science Certificate (In progress)

■ PROFESSIONAL EXPERIENCE

University of Iowa

Assistant Professor, Science Education, Department of Teaching and Learning (2022-Present)

University of Texas at Austin

Researcher, Center for Community College Student Engagement, 2021-2022

Arkansas Tech University

Assistant Professor, Science Education, Department of Physical Sciences, 2019-2021

New York University

Visiting Scholar, Center for K-12 STEM Education, Tandon School of Engineering, 2018-2019

Michigan State University

Research Associate, Create for STEM Institute, 2017

New York University

Research Associate, Department of Mechanical and Aerospace Engineering, 2016-2017

University of Texas at Austin

Research Assistant, Departments of Educational Psychology, Chemical Engineering, Geological Sciences, Curriculum & Instruction, and School of Biological Sciences (various roles), 2010-2021

Teaching Assistant, Departments of Educational Psychology, Curriculum & Instruction, and College of Natural Sciences, 2012-2021

Seoul Metropolitan City, South Korea

Science Teacher, Shin Myeong Public Middle School (2008-2010); Dolma Public High School (2007)

■ RESEARCH INTERESTS

Assessment Development and Validation

Technology-integrated Instructional Practices

AI Education in K-16

Learning Analytics

■ PUBLICATIONS

You, H. (2016). Toward Interdisciplinary Science Learning: Development of an Assessment for Interdisciplinary Understanding of Carbon Cycling [Doctoral dissertation, University of Texas at Austin]. ProQuest Dissertations & Theses Global.

You, H. (2008). Bridging the Gap Between High School and College: Exploration of Vertical Curriculum Coherence Focusing on Carbon Compounds [Unpublished master's thesis, Yonsei University]

Refereed Journal Articles (27)

You, H., Nordine, J., Maeng, S., Tanus, J., Kaufmann, L., Hatcher, K., & Onwukeme, I. (2026). Developing a Conceptual Framework of Interdisciplinary Understanding for Meaningful Learning. *Educational Sciences*. (**Educational Sciences** is a Q1 journal in the *Education and Educational Research* category, with a 2025 impact factor of 3.5 and a five-year impact factor of 3.3).

You, H., Park, S., Delgado, C., & Yang, S. (2026). Advancing Interdisciplinary Science Education: Validating the Interdisciplinary Science Assessment of Carbon Cycling II through Internal Structure and External Factors. *Research in Science Education*. (**RISE** is a top-tier science education research journal. 2022 ISI Journal Citation Reports impact factor = 2.3, Indexed in SSCI, SCOPUS etc.)

You, H. & Hong, M. (2026). Next-Gen Assessment for Interdisciplinary Learning: Development and Validation of the Interdisciplinary Science Assessment of Carbon Cycling II. *International Journal of Science Education*. (**IJSE** is indexed in SSCI and Scopus, with a 2024 Journal Impact Factor of 2.3, a five-year Impact Factor of 3.3, and a 2024 CiteScore of 5.0)

Harper-Gampp, T., Delgado, C., **You, H.**, Peterson M., & Chen, K. (2025). Measuring size and scale: The development and validation of the Assessment of Size and Scale Cognition (ASSC). *Research in Science Education*, 1-24. (**RISE** is a top-tier science education research journal. 2022 ISI Journal Citation Reports impact factor = 2.3, Indexed in SSCI, SCOPUS etc.)

Zhu, L., **You, H.**, Hong, M., & Fang, Z. (2025). Predictive Insights into U.S. Students' Mathematics Performance in PISA 2022: Using Ensemble Tree-Based Machine Learning Models. *International Journal of Educational Research*, 130, 102537. (**IJER** is a SSCI- and Scopus-indexed journal with a 2022 impact factor of 2.6 and a SCImago Journal Rank (SJR) of 0.969 (Q1))

You, H., Hong, M., Zhu, L., & Fang, Z. (2025). What predicts scientific literacy of U.S. students: revealing influential factors and group comparisons through the application of a supervised machine learning model. *The International Journal of Science and Math Education*, 1-29. (**IJSME's** impact factor is 2.1 in 2024, Indexed in SSCI, SCOPUS etc.)

You, H., Park, S., Hong, M., Warren, A. (2024). A meta-analysis examining the impact of professional development programs in science education. *Journal of Research in Science Teaching*, 62(4), 971-1005. (**JRST** is the top-ranked science education research journal. Rank 49/270 education and education research journals. 2026 Impact Factor= 4.5, Journal Citation Indicator of

2.05, a CiteScore of 8.1, 12.5% acceptance rate)

Hong, M. & You, H. (2024). Enhancing rubric development in science education through topic modeling techniques. *The Journal of Experimental Education*, 1-18.
(2023 ISI Journal Citation Reports impact factor = 2.9, Indexed in SSCI, SCOPUS etc. 16% acceptance rate)

You, H. & Hong, M. (2024). Looking beyond disciplinary silos: Exploring students' interdisciplinary science understanding using topic modeling. *Research in Science Education*, 54(4), 1-25.
(RISE is a top-tier science education research journal. 2022 ISI Journal Citation Reports impact factor = 2.3, Indexed in SSCI, SCOPUS etc.)

Sripathi, K., Moscarella, R., Prevost, L., You, H., Yoho, R., Merrill, J., Haudek, K., & Urban-Lurain, M. (2023). Machine learning mixed methods text analysis: An illustration from automated scoring models of student writing in biology education. *Journal of Mixed Methods Research*, 18(1), 48-70.
(JMMR is a peer-reviewed international journal that focuses on empirical methodological articles. 2021-2022 ISI Journal Citation Reports IF = 5.746, Indexed in SSCI, Scopus etc.)

You, H., Park, S., Marshall, J., & Delgado, C. (2022). Interdisciplinary science assessment of carbon cycling: Construct validity evidence based on internal structure. *Research in Science Education*, 52, 473-492.
(RISE is a top-tier science education research journal. 2022 ISI Journal Citation Reports impact factor = 2.3, Indexed in SSCI, SCOPUS etc.)

You, H. (2022). Bayesian versus frequentist estimation for Item Response Theory (IRT) models of interdisciplinary science assessment. *Interdisciplinary Journal of Environmental and Science Education*, 18(4), e2297.
(IJESE is peer-review international science education journal, acceptance rate in 2021 = 23%)

You, H., Marshall, J., & Delgado, C. (2021). Toward interdisciplinary learning: Development and validation of an assessment for interdisciplinary understanding of global carbon cycling. *Research in Science Education*, 51, 1197-1221 .
(RISE is a top-tier science education research journal. 2022 ISI Journal Citation Reports impact factor = 2.3, Indexed in SSCI, SCOPUS etc.)

You, H., Delgado, C., & DeAtley, K. (2021). Experts' model-based reasoning and interdisciplinary understanding of carbon cycling, *International Journal of Research in Education and Science*, 7(2), 562-579.
(IJRES is a peer-reviewed international science and education journal. 2019 SJR = 0.19, acceptance rate in 2021 = 12%)

You, H., & Mary, S., & Kapila, V. (2021). Effectiveness of professional development: Integration of educational robotics into science and mathematics curricula. *Journal of Science Education and Technology*, 30, 567-581.
(JOSET is a top-tier, peer-reviewed science and technology education journal. 2021 ISI IF = 3.419, Indexed in SSCI and Scopus etc.)

You, H., Park, S., & Delgado, C. (2020). A closer look at U.S. schools: What characteristics are associated with scientific literacy? A multivariate multilevel analysis using PISA 2015. *Science Education*, 105(2), 406-437.
(SciEdu is the top-ranked science education research journal. Rank 14/270 education and education research journals. 2021 ISI Journal Citation Reports IF = 6.000, indexed in Scopus,

SSCI etc.)

- Sripathi, K., Moscarella, R., Nehm, R., Yoho, R., **You, H.**, Urban-Lurain, M., Haudek, K., & Merrill, J. (2019). Mixed student ideas about tracing matter across biological scales in the context of human weight loss. *CBE—Life Sciences Education*, 18(3), 1-17.
(LSE is a peer-reviewed articles on life science education and is published by the American Society for Cell Biology (ASCB), 2017 ISI Journal Citation Reports impact factor = 4.10, h-index = 74, SJR= 1.297, Indexed in SCIE and PubMed)
- You, H.**, Marshall, J., & Delgado, C. (2018). Assessing students' disciplinary and interdisciplinary understanding of global carbon cycling. *Journal of Research in Science Teaching*, 55(3), 377-398.
(JRST is the top-ranked science education research journal. Rank 49/270 education and education research journals. 2026 Impact Factor= 4.5, Journal Citation Indicator of 2.05, a CiteScore of 8.1, 12.5% acceptance rate)
- You, H.**, Kim, K., Black, K. A., & Min K. (2018). Assessing science motivation for college students: Validation of the Science Motivation Questionnaire II using the Rasch-Andrich Rating Scale Model. *EURASIA Journal of Mathematics, Science and Technology Education*, 14(4), 1161-1173.
(EJMSTE is a peer-reviewed and scholarly journal. ISI Journal Citation Reports = 4.4, SJR = 0.569, Indexed in SSCI etc.)
- Marshall, J., Banner, J., & **You, H.** (2018). Assessing the effectiveness of sustainability learning. *Journal of College Science Teaching*, 47(3), 57-67.
(JCST is a peer-reviewed journal published by the National Science Teachers Association)
- Delgado, C., Jones, M. G., **You, H.**, Roberston, R., & Halberda, J. (2017). Scale and the evolutionarily-based approximate number system: An exploratory study. *International Journal of Science Education*, 39(8), 1008-1024.
(IJSE is a top-tier science education research journal. 2021 ISI Journal Citation Reports impact factor = 3.00 rank 104/ 219 education and education research journals, SJR= 1.151, h-index= 115, indexed in Scopus, SSCI etc)
- You, H.** (2017). Why teach science with an interdisciplinary approach?: History, trends, and conceptual frameworks. *Journal of Education and Learning*, 6(4), 66-77.
(JEL is an open-access, international, double-blind peer-reviewed journal published by the Canadian Center of Science and Education, 2021 Google-based impact factor = 1.93, h-index =48)
- You, H.** (2016). Rasch validation of reform-oriented science teaching practice scale. *Journal of Science Teacher Education*, 27(4), 373-392.
(JSTE is a top-ranked science teacher education journal, 2023 cite score = 2.1 from Scopus, SJR = 0.866)
- You, H.** (2016). No teacher left behind: Curriculum knowledge of science teachers. *Future of Education Research*, 29(2), 1-22.
(FER is published by the institute for Educational Research at Yonsei University, Indexed in KCI)
- You, H.** (2015). Do schools make a difference?: Exploring school effects mathematics achievement in PISA 2012 using hierarchical linear modeling. *Journal of Educational Evaluation*, 28(5), 1301-1327.
(JEE is a top-ranked educational measurement and evaluation journal in South Korea,

You, H., & Delgado, C. (2014). Toward an interdisciplinary science curriculum: Analysis of the connections across science learning progressions. *International Journal for Cross-Disciplinary Subjects in Education*, 4(1), 1854-1862.
(IJCDSE is a peer-reviewed, open access journal, published quarterly. Indexing Citation Board IF = 5.214 calculated)

Manuscript and Book Chapter In Revision And Review (7)

You, H. & Zhu, L. (under review with major revision). Developing a Custom GPT for NGSS-PLUS-5E Lesson Planning in Elementary Science Education. *Journal of College Science Teaching*.

You, H., Park, S., Zhu, L., Hong, D., Hong, M., & Jung, J. (under review with major revision). Advancing Professional Development for K–12 Math Educators: Meta-Analysis of Effects, Challenges, and Next Steps, *International Journal of Science and Mathematics Education*.

You, H., Park, S., & Hong, M. (under review). Teacher Professional Development in STEM Education (2015–2025): A Meta-Analysis of Effects and Emerging Directions. *Humanities and Social Sciences Communications*.

You, H., Hong, M., & Maeng, S. (under review). Characterizing College Students' Interdisciplinary Understanding of Carbon Cycling Using Supervised Latent Dirichlet Allocation. *Science Education*.

Zhu, L., You, H., Fang, Z., & Hong, D. (under review). U.S. Rural Students' Mathematics Performance: A Transfer Learning Study of PISA 2022. *Computer & Education*.

Choi, S., & You, H. (under review). Navigating Students' Challenges and Solutions in VR-Based Science Lessons. *Computer & Education*

Bae, Y., Kim, E., & You, H. (under review). The Development and Validation of Teacher Beliefs about Multicultural Education (TBME) Questionnaire. *Journal of Multicultural Education*.

Referred Book Chapter (5)

Wagner, L, Won, H., & You, H. (in press). AI-driven approaches in early childhood and elementary education: Insights from methods courses in three teacher education programs. In E. Jones, B. Crawford, & M. Clay (Eds.), *Embracing artificial intelligence in the classroom*, Information Age Publishing.

You, H., & Zhu, L. (in press). Inclusiveness and assessments. In A. Ben (Ed.), *Springer Nature*.

You, H. (2022). Revisiting the relationship between science teaching practice and scientific literacy: Multi-level analysis using PISA 2015. In M. S. Khine (Ed.), *Multi-level modeling*, Springer.

Won, H. & You, H. (2022). Next generation science and engineering teaching practices in a pre-K classroom. *Early childhood education - Innovative pedagogical approaches in the post-modern era*, IntechOpen.

You, H., Haudek, K., Merrill, J., & Urban-Lurain, M. (2020). Construct validity of computer scored constructed response items in undergraduate introductory biology courses. In M. S. Khine (Ed.), *Rasch measurement: Applications quantitative educational research*, Springer.

Refereed Proceedings (9)

- Randall, D. & **You, H.** (2024). Bridging Rhetoric and Engineering: Qualitative Results from a Writing Center Program to Improve Engineering Undergraduate Writing, IEEE International Professional Communication Conference (ProComm), Pittsburgh, pp. 145-152, doi: 10.1109/ProComm61427.2024.00034.
- Randall, D & **You, H.**, Puperi, D., Lindsey, T., & Brooke, R. (2023). Write from the start in engineering: Mixed-methods results of a collaboration between a first-year biomedical engineering class and a university writing center. Proceedings of American Society for Engineering Education (ASEE). <https://peer.asee.org/44440>.
- You, H.** & Hong, M. (2022). Looking beyond disciplinary silos: Students' interdisciplinary science understanding of carbon cycling. In Chinn, C., Tan, E., Chan, C., & Kali, Y. (Eds.), Proceedings of the 16th International Conference of the Learning Sciences - ICLS 2022 (pp. 1832-1833). International Society of the Learning Sciences (ISLS).
- You, H.**, Mary, S., Borges, S. I., & Kapila, V. (2019). Designing robotics-based science lessons aligned with the three dimensions of NGSS-plus-5E Model: A case study of middle school teachers. Proceedings of American Society for Engineering Education (ASEE).
- You, H.**, Mary, S., & Kapila, V. (2019). Teaching science with technology: Scientific and engineering practices of middle school science teachers engaged in a robot-integrated professional development program. Proceedings of American Society for Engineering Education (ASEE).
- You, H.**, & Kapila, V. (2017). Effectiveness of professional development: Integration of educational robotics into science and math curricula. Proceedings of American Society for Engineering Education (ASEE).
- You, H.**, Park, S., Marshall, A. J., & Maeng, S. (2017). Internal structure and fairness of the interdisciplinary science assessment. Proceedings of the European Science Education Research Association (ESERA), Dublin, Ireland.
- You, H.**, & Delgado, C. (2014). Weaving an interdisciplinary science curriculum: Analysis of the connections across learning progressions. In C. A. Shoniregun & G. A. Akmayeva (Eds.), Proceedings of Canada International Conference on Education Conference (pp. 68-71). Basildon, UK: CICE.
- Zeeuw, A. D., Craig, T., & **You, H.** (2013). Assessing conceptual understanding in mathematics. IEEE Frontiers in Education Conference Proceedings (pp.1742-1744), Oklahoma City, OK.

Work In Progress (5)

- You, H.**, Fang, Z., & Kaffuman, L.. (in preparation). AI-Enhanced Feedback as a Driver of Meaningful Learning in Carbon Cycling, International Journal of Science Education..
- You, H.**, Zhu, L., Moon, J., Aldemir, T., Lee, H., Song, D., Abu, S., Emerson, W., & Wang, S. (in preparation). How Technology is Shaping K-12 Science Learning: A Scoping Review. Will be submitted to JRST in 2026.
- You, H.**, Zhu, L., Moon, J., Aldemir, T., Lee, H., Song, D., & Abu, S. (in preparation). Virtual Reality and Augmented Reality in K-12 Science Education: A Systematic Review of Applications and Impacts. Will be submitted to the Journal of Science Education and Technology in 2026.

You, H. & Park, S. (in preparation). Modeling Teachers' and Students' Cognitive and Affective Outcomes From Teacher Professional Development: Evidence From META-SEM. *Journal of Science Teacher Education* in 2027.

■ PRESENTATIONS

Refereed Presentations (64)

Add two submitteds

You, H., Fang, Z., & Kauffman, L. (submitted). AI-Enhanced Feedback as a Driver of Interdisciplinary Understanding in Carbon Cycling, National Association for Research in Science Teaching (NARST), Boston, MA.

You, H., Hong, M., & Maeng, S. (submitted). Characterizing College Students' Interdisciplinary Understanding of Carbon Cycling Using Supervised Latent Dirichlet Allocation, National Association for Research in Science Teaching (NARST), Boston, MA.

You, H. & Lee, H. (submitted). Assessing Preservice Teachers' Interdisciplinary Scientific Understanding, Systems Thinking, and Sustainability Perspectives, National Association for Research in Science Teaching (NARST), Boston, MA.

You, H., Park, S., & Zhu, L. (2026, July). Understanding the Pathways of Professional Development: Teacher Change and Student Outcomes Across Cognitive and Affective Domains, Korean Association for Science Education (KASE), Korea.

You, H. (2026, June). Designing Meaningful and Personalized Interdisciplinary Science Learning with AI Feedback, International Conference on Learning Sciences and Educational Innovation (ICOLSEI), Korea.

Hong, M. & You, H. (2026, Apr). Differential Item Functioning Detection Methods in Mixed-Format Assessments Using Propensity Score Weighting, National Council on Measurement in Education (NCME), LA, CA.

You, H., Park, S., Zhu, L., Hong, D., & Hong, M. (2026, Apr). Advancing Professional Development for K–12 Math Educators: Meta-Analysis of Effects, Challenges, and Next Steps, American Educational Research Association (AERA), LA, CA.

You, H. & Hong, M. (2026, Apr). Uncovering College Students' Epistemological Foundations of Interdisciplinary Understanding of Carbon Cycling Through Constructed Responses, American Educational Research Association (AERA), LA, CA.

You, H. & Zhu, L. (2026, Apr). Understanding U.S. Rural Students' Mathematics Performance: A Transfer Learning Approach on PISA 2022, American Educational Research Association (AERA), LA, CA.

Mentesoglu, Z. & You, H. (2026, Apr). Predicting Science Achievement: A Machine Learning Approach to PISA Data Analysis Across 2006 and 2015, American Educational Research Association (AERA), LA, CA.

You, H., Fang, Z., & Zhu, L. (2026, Apr). ISACC II meets ChatGPT: Enhancing interdisciplinary science learning through AI-driven personalized feedback, National Association for Research in Science Teaching (NARST), Seattle, WA.

- You, H., & Park, S.** (2026, Apr). From Structure to Fairness: Validating the Interdisciplinary Science Assessment of Carbon Cycling II, National Association for Research in Science Teaching (NARST), Seattle, WA.
- You, H., Hong, M., & Lee, W.** (2026, Apr). Bridging Disciplinary Boundaries: Development and Validation of the Interdisciplinary Science Assessment of Carbon Cycling II, National Association for Research in Science Teaching (NARST), Seattle, WA.
- You, H., & Park, S.** (2025, July). *Improving teaching practices, enhancing learning outcomes: A meta-analysis of science professional development impact on teachers and students.* Korean Association for Science Education (KASE), Korea.
- You, H.** (2025, June). AI-Powered Feedback for Constructed Response Science Items: Improving Teacher Efficiency, Instructional Practices, and Student Learning Outcomes, The Paris Conference on Education, Paris, France.
- You, H., Park, S., & Yang, S.** (2025, Mar). Creating and Validating the Interdisciplinary Science Assessment of Carbon Cycling II, National Association for Research in Science Teaching (NARST), Washington, DC.
- You, H., Hong, M., & Maeng, S.** (2025, Mar). Unlocking Interdisciplinary Insights and Understandings on Carbon Cycling through Topic Modeling, National Association for Research in Science Teaching (NARST), Washington, DC.
- Zhu, L., You, H., Hong, M., & Fang, Z.** (2025, April). Predictive Insights into U.S. Students' Mathematics Performance in PISA 2022: Using Ensemble Tree-Based Machine Learning Models. American Educational Research Association (AERA), Denver.
- D. Randall & H. You.** (2024, July), Bridging Rhetoric and Engineering: Qualitative Results from a Writing Center Program to Improve Engineering Undergraduate Writing, IEEE International Professional Communication Conference (ProComm), Pittsburgh.
- You, H., Hong, M., Maeng, S., & Park, S.** (2024, June). Unveiling Students' Interdisciplinary Understanding of Carbon Cycling: Employing a Topic Modeling, Korean Educational Research Association, Korea
- Hong, M., You, H., & Duffey, M.** (2024, April). Enhancing Assessment Rubrics: Unraveling Interdisciplinary Science Understanding through Topic Modeling Techniques. American Educational Research Association (AERA), Philadelphia.
- You, H., Hong, M., Zhu, L., & Fang, Z.** (2024, March). What Predicts Scientific Literacy of U.S. Students: Revealing Influential Factors and Group Comparisons through the Application of a Supervised Machine Learning Model. National Association for Research in Science Teaching (NARST), Denver.
- You, H., Park, S., Hong, M., & Warren, A.** (2024, March). Untangling the Effects: A Meta-Analysis Examining the Impact of Professional Development Programs For K-12 Science Teachers. National Association for Research in Science Teaching (NARST), Denver.
- You, H., Zhu, L., Won, H., & Wagner, L.** (2023, October). Enhancing Teaching Efficiency and Personalizing Learning: The Advantages of ChatGPT for Teachers' Lesson Planning. National Science Teacher Association (NSTA), Kansas City.
- Randall, D., You, H., Puperi, D., Lindsey, T., & Brooke, R.** (2023, June). Write from the Start in Engineering: Mixed-Methods Results of a Collaboration Between a First-Year Biomedical Engineering Class and A University Writing Center. American Society for Engineering

Education (ASEE), Baltimore.

You, H., & Hong, M. (2023, July). Unlocking Interdisciplinary Perspectives: Exploring Students' Understanding of Carbon Cycling Using Topic Modeling. Korean Association for Science Education (KASE), Korea.

You, H., Park, S., & Hong, M. (2023, April). Effectiveness of Professional Development for STEM Teachers on Teachers and Students: Meta-Analysis. American Educational Research Association (AERA), Chicago.

Hong, M., & You, H. (2023, April). Revealing Students' Interdisciplinary Understanding of Carbon Cycling Using the Topic Modeling Technique. American Educational Research Association (AERA), Chicago.

Randall, D. & You, H. (2023, February). Engineering Sentences through the Texas Snowpocalypse: Results of a Collaboration between a University Writing Center and an Engineering Writing Course. Conference on College Composition & Communication, Chicago.

You, H. & Hong, M. (2022, June). Looking Beyond Disciplinary Silos: Students' Interdisciplinary Science Understanding of Carbon Cycling. The International Conference of the Learning Sciences (ICLS), Hiroshima, Japan.

You, H., Park, S., & Hong, M. (2022, March). What Works in K-12 STEM Professional Development Programs?: A Meta-Analysis of Its Impacts on Teachers and Students. The National Association for Research in Science Teaching (NARST), British Columbia, Vancouver.

You, H., & Lee, S. (2022, March). Bayesian Versus Frequentist Estimation for Item Response Theory (IRT) Models of Interdisciplinary Science Assessment. National Association for Research in Science Teaching (NARST), British Columbia, Vancouver.

Hong, M., & You, H. (2021, April). Analysis Of Students' Written Answers in Interdisciplinary Science Assessment: Application of Topic Modeling Technique. Proposal Presented at the Division H 2021 Early Career/Graduate Student In-Progress Research Roundtable of American Educational Research Association (AERA) Annual Meeting.

You, H., Park, S., & Hong, M. (2021, April). Professional Development Effectiveness in STEM Education: Meta-Analysis. National Association for Research in Science Teaching (NARST).
You, H., & Park, S. (2021, April). Revisiting the Relationship between Science Teaching Practice and Scientific Literacy from Global Perspective. National Association for Research in Science Teaching (NARST).

Delgado, C., You, H., & Murrillo, N. (2021, April). Analysis of the Spanish-Language Force Concept Inventory: Lost in Translation? National Association for Research in Science Teaching (NARST).

You, H., Delgado, C., & Deatley, K. (2021, April). Experts' Model- Based Reasoning and Interdisciplinary Understanding about Carbon Cycling. American Educational Research Association (AERA).

You, H., Park, S., & Delgado, C. (2021, April). A Closer Look at U.S. Schools: What School Characteristics are Associated with Scientific Literacy? A Multivariate Multilevel Analysis Using PISA 2015. American Educational Research Association (AERA), Orlando.

You, H. & Won, H. (2020, April). Exploring Global Science Teaching Practices from the PISA

- Perspective. American Educational Research Association (AERA), San Francisco. You, H., Delgado, C., & Park, S. (2020, April). Exploring Differential School Effects between Low and High Ability Groups on Scientific Literacy. American Educational Research Association (AERA), San Francisco.
- Won, H., Jones, L., Condo J., Huh, I., & **You, H.** (2020, April). STEM Teaching and Learning in Preschool Setting: A Systematic Review of Empirical Research (2009-2018). American Educational Research Association (AERA), San Francisco.
- Won, H., Jones, L., **You, H.**, & Condo, J. (2019, February). A Case Study: Investigating a Preschool Teacher's Beliefs and Practices in STEM Teaching and Learning. Eastern Educational Research Association (EERA), Myrtle Beach.
- You, H.**, Mary, S., Borges, S. I., & Kapila, V. (2019, June). Designing Robotics-Based Science Lessons Aligned with the Three Dimensions of NGSS-Plus-5E Model: A Case Study of Middle School Teachers. American Society for Engineering Education (ASEE), Tampa.
- You, H.**, Mary, S., & Kapila, V. (2019, June). Teaching Science with Technology: Scientific and Engineering Practices of Middle School Science Teachers Engaged in a Robot-Integrated Professional Development Program. American Society For Engineering Education (ASEE), Tampa.
- You, H.**, Park, S., & Delgado, C. (2019, April). Toward Excellence with Equity: School Effects on Scientific Literacy in PISA 2015. American Educational Research Association (AERA), Toronto, Canada.
- You, H.**, Merrill, J., Haudek, K., & Urban-Lurain, M. (2019, March). Rasch Validation of Computer Scored Constructed Response Items in Undergraduate Introductory Biology Courses. National Association for Research in Science Teaching (NARST), Baltimore.
- Won, H., Song, Y., & **You, H.** (2019, April). NGSS-Based Instructional Practices in Pre-K STEM Teaching. American Educational Research Association (AERA), Toronto, Canada.
- Won, H., Jones, L., Pan, Z., **You, H.**, & Puckett, K. (2019, April). Teacher's Educational Beliefs in Shaping Instructional Practices for Pre-K's STEM Learning. American Educational Research Association (AERA), Toronto, Canada.
- Sripathi, K., Moscarella, R., Yoho, R., **You, H.**, Nehm, R., Urban-Lurain, M., Haudek, K., & Merrill, J. (2018, July). Mixed Students' Ideas about Tracing Matter across Biological Scales in the Context of Human Weight Loss. Society For The Advancement Of Biology Education Research (SABER), Twin Cities.
- Song, Y., **You, H.**, Martin-Hanses, & Gomez-Zwiep. (2018, March). The Impact of the Course on College Students' Spatial Thinking Abilities. National Association for Research in Science Teaching (NARST), Atlanta.
- You, H.**, Marshall, J., & Delgado, C. (2018, March). Assessing Students' Disciplinary and Interdisciplinary Understanding of Global Carbon Cycling. National Association for Research in Science Teaching (NARST), Atlanta.
- Delgado, C., & **You, H.** (2018, April). Interdisciplinary Connections in the NGSS: Realizing the Vision. American Educational Research Association (AERA), New York.
- You, H.**, Park, S., & Marshall, A. J. (2017, August). Internal Structure and Fairness of the Interdisciplinary Science Assessment. The European Science Education Research

Association (ESERA), Dublin, Ireland.

- You, H.**, Sripathi, K., Yoho, R., Moscarella, R., Merrill, J., Haudek, K., & Urban-Lurain, M. (2017, July). Towards Meaningful Scores: Development and Validation of Analytic Rubrics for Diagnostic Purposes. Society for the Advancement of Biology Education Research (SABER), Twin Cities.
- Marshall, A. J., Banner, J., & **You, H.** (2017, July). The Interaction between Physics Learning and Interdisciplinary Learning. American Association of Physics Teachers (AAPT), Cincinnati.
- You, H.**, & Kapila, V. (2017, June). Effectiveness of Professional Development: Integration of Educational Robotics into Science and Math Curricula. American Society for Engineering Education (ASEE), Columbus.
- Song, Y., Martin-Hanses, Gomez-Zwiep, & **You, H.** (2017, April). The Impact of the Innovative Course on Developing Spatial Thinking Abilities in College Students. National Association for Research in Science Teaching (NARST), San Antonio.
- You, H.**, Marshall, A. J., & Delgado, C. (2017, April). Toward Interdisciplinary Science Learning: Development of an Assessment for Interdisciplinary Understanding of 'Carbon Cycling'. National Association for Research in Science Teaching (NARST), San Antonio.
- You, H.**, & Sathasivan, K. (2016, July). A Positive Impact of a New Classroom Response System Called Squarecap on Student Grades and Instructor Evaluation. The American Society of Plant Biologists (ASPB), Austin.
- You, H.**, Kim, K., Black, K. A., & Min K. (2016, April). Validation of the Science Motivation Questionnaire II Using Classical Test Theory and Rasch-Andrich Rating Scale Model. National Association for Research in Science Teaching (NARST), Baltimore.
- Hwang, Y., **You, H.**, & Park, Y. (2015, Oct). Extraction and Verification of Problem Specification Elements Used in Research Procedure of Science and Engineering. 2015 International Conference of East-Asian Association for Science Education, Beijing, China.
- You, H.** (2015, May). Rasch Calibration Of Measuring Math Teachers' Use of Cognitive-Activation Strategies. 2015 Chicago International Conference On Education (ICE), Chicago.
- You, H.** (2015, April). Rasch Calibration: Reform-Oriented Teaching Practices. American Educational Research Association (AERA), Chicago.
- You, H.**, & Delgado, C. (2015, April). Using Crosscutting Concepts as a Vehicle for the Design of Interdisciplinary Assessments. In J. Shen (Chair), Interdisciplinary and Integrated STEM Education: Research, Practices, and Perspectives. Symposium Conducted at the Meeting of the American Educational Research Association, Chicago.
- You, H.**, & Delgado, C. (2015, April). Revisiting The Coleman Report: Exploring School Effects on Scientific Literacy in PISA 2012 Using Hierarchical Linear Modeling. National Association for Research in Science Teaching (NARST), Chicago.
- You, H.** (2014, June). Weaving an Interdisciplinary Science Curriculum: Analysis of the Connections Across Learning Progressions. The Canada International Conference on Education (CICE), Sydney, Nova Scotia, Canada.
- You, H.** (2014, January). Development and Validation for Measuring Teachers' Curriculum Knowledge. Hawaii International Conference on Education, Honolulu.
- You, H.** (2014, Jan). The Widening Achievement Gap by Socioeconomic Status. Hawaii International

Conference on Education, Honolulu.

Zeeuw, A. D., Craig, T., & **You, H.** (2013, October). Assessing Conceptual Understanding in Mathematics. *Frontiers in Education*, Oklahoma City.

Delgado, C. & **You, H.** (2012, April). Size And Scale Tasks and their Relation to Evolutionarily-Based and Culturally-Based Knowledge. *The National Association for Research in Science Teaching*, Rio Grande, Puerto Rico.

Delgado, C. & **You, H.** (2011, March). Theoretical and Empirical Investigation of Students' Strategies for Size Estimation. *The National Association for Research in Science Teaching*, Indianapolis.

■ GRANTS, AWARDS, AND FELLOWSHIPS

2026

You, H. (PI). Chen, Z (Co-PI), Lee, Y. (Co-PI), & Mcdermott M. (Co-Pi). AI-Personalized Feedback to Promote Interdisciplinary Science Understanding and Systems Thinking Toward Meaningful Learning, Digital Promise (\$250K, Pending).

You, H. (PI) & Chen, Z (Co-PI). AI-Powered Formative Feedback to Enhance Interdisciplinary Science Understanding and Systems Thinking in K–12 Education, Proposal representing UI, NSF (\$300K, Pending).

2025

You, H. (PI). Chen, Z (Co-PI). Improving Math Assessment Accessibility for Early Elementary Students Through Minimized Reading Demands, AIMS EduData Initiative (\$250K, Not Selected).

You, H. (PI). AI-Enhanced Feedback as a Driver of Meaningful Learning in Science Education, Spencer Small Grant (\$50,000, Not Selected).

You. H. The Role of Artificial Intelligence and Machine Learning in K–12 Science Education: A Systematic Review, Global Research Partnerships Award, University Of Iowa (\$10,000)

Seo. H. (PI), **You. H. (Co-PI)**, LeFevre, G (Co-I), & Mattes, T (Co-I). RET Site: Teachers' Research Initiative in Engineering of Biology and Manufacturing at the University of Iowa (\$600K, Submitted)

You, H. (PI). Generative AI in Science Education: Effectiveness, Comparisons, and Student Perceptions of Feedback, Iowa Initiative for Artificial Intelligence, University of Iowa (\$25,000, Not Selected)

You. H. (PI) & Seo H. (Co-PI). Obermann Interdisciplinary Research Grants, University of Iowa (\$3000, Not Selected)

You. H. (PI). Teaching & Learning Summer Graduate Research Assistant Award, University of Iowa (\$1221)

You. H. (PI). Shaping the Future of K–12 AI Education: Insights from Korea, Singapore, and the United States, Summer Research Fellowship Award, University of Iowa (\$3,000)

You. H. (PI). A Meta-SEM Analysis of Technology Integration and Learning Outcomes in K–12 Science Education. International Travel Award, University of Iowa (\$1,000)

You. H. (PI). Artificial Intelligence–Based Personalized Feedback. Investment in Strategic Priorities,

University of Iowa (\$5,000)

2024

You. H. Meritorious Research Award, College of Education, University of Iowa

You, H. (PI). Predictive Insights into U.S. Students' Mathematics and Science Performance on PISA Using Ensemble Tree-Based Machine Learning Models, Iowa Initiative for Artificial Intelligence, University of Iowa (\$25,000)

You. H. (PI). Curriculum Development for Sustainability-Focused General Education, ASEE Engineering for One Planet Mini-Grant Program (EOP-MGP) (\$8000, Not Selected)

Park. H. (Lead PI), **You. H. (PI)**, & Kang, T. (PI). Collaborative Research: Inclusive and Dynamic STEM Learning through AI-Enhanced Video Search with Adaptive Mind Mapping, NSF RITEL PROGRAM (\$900,000, Not Selected)

Beckstead. J. (PI), **You. H. (Co-I)**, Won. H. (Co-I). Mars Area Readiness for Space (MARS) (250,000, Submitted)

You. H. (PI) & Park. S. (Co-PI). STEM Educator Growth: Insights from a Meta-Analysis and Future Pathways, Spencer Foundation (\$125,000, Not selected)

Lee. Y. (Lead PI), **You. H. (PI)**, Metzger., J (PI), & Sun F. (Co-PI). Transforming Inclusive and Authentic STEM Education through AI and Immersive Technologies. (TranSTEM). NSF RITEL PROGRAM (\$900,000, Not Selected)

Chen. S. (Lead PI), **You. H. (PI)**, Lee. Y. (Co-PI), Shyu M., J (Co-PI). Collaborative Research: DSC: NextGenDS - Building Next Generation Data Science Training and Education Pathways. NSF RITEL (\$900,000, Not Selected)

You. H. (PI). How Technology is Shaping K-12 Science Learning: A Scoping Review. Global Research Partnerships Award, University Of Iowa (\$10,000, Not Selected)

You. H. (PI) & Won H. (Co-PI). Obermann Interdisciplinary Research Grants, University of Iowa (\$3000, Not Selected)

You. H. (PI). Obermann Center Working Group Grants, University of Iowa (\$500)

You. H. (PI). Unveiling Students' Interdisciplinary Understanding of Carbon Cycling Using Topic Modeling, Summer Research Fellowship Award, University of Iowa (\$3,000)

You. H. (PI). A Meta-Sem Analysis on the Effectiveness of Professional Development Programs for STEM Teachers. International Travel Award, University of Iowa (\$1,000)

You. H. (PI). Teaching & Learning Summer Graduate Research Assistant Award, University of Iowa (\$1,000)

2023

Zawily, A. (PI), **You. H. (Co-PI)**, & Mcgrane, R. (Co-PI). Catalyzing Excellence: Promoting Student Persistence through Collaborative Cures with Institutional Connections. NSF IUSE (\$400,000, Not Selected)

You. H. (PI) & Park. S. (Co-PI). STEM Educator Professional Development: A Meta-Analysis and Future Directions, Spencer Foundation (\$60,000, Not Selected)

- Lee. Y. (Lead PI), **You. H. (PI)**, Metzger., J (PI), & Sun F. (Co-PI). Transforming Inclusive and Authentic STEM Education through AI and Immersive Technologies. (TranSTEM). NSF RITEL (\$900,000, Not Selected)
- You. H. (PI)**. Decadal Insights: A Meta-Analysis Examining the Impact of Professional Development Programs in Science Education. Old Gold Summer Fellowship, University of Iowa (\$6,000)
- You. H. (PI)**. Does Technology Matter? Exploring The Impact Of Virtual Reality-Enhanced Project-Based Learning on Student Learning Outcomes. Summer Research
- You. H.** Fellowship Award, University of Iowa (\$3,000)
- You. H. (PI)**. Teaching & Learning Summer Graduate Research Assistant Award, University of Iowa (\$1,150)
- You. H. (PI)**. Learning Together: Motivation and Attitudes Towards Learning Science in Multicultural Classrooms. International Travel Award, University of Iowa (\$1,200)
- You. H. (PI)**. Elementary Preservice Teachers' Interdisciplinary Science Understanding of Carbon Cycling. Global Research Partnerships Award. University of Iowa (\$10,000, Not Selected)
- You. H. (PI)**. What Predicts Scientific Literacy Of U.S. Students: Revealing Influential Factors and Group Comparisons through the Application of a Supervised Machine Learning Model. Funding For Investment In Strategic Priorities, University of Iowa (\$5,000)
- You. H. (PI)**. Student Tech Fee for Integrating educational robotics into Science Methods I, University of Iowa (\$1999.75)
- You. H. (PI)**. Development and Validation of a Survey for Measuring Mental Health-Related Quality of Life (MHRQOL) for K-12 Students, Scanlan Center for School Mental Health Grant, University of Iowa (Not Selected)

2022

- You. H. (PI)**. Elementary Preservice Teachers' Interdisciplinary Science Understanding of Carbon Cycling. 2022-2023 Global Research Partnerships Award, University Of Iowa (\$10,000)
- You. H. (PI)**. Unlocking Interdisciplinary Perspectives: Exploring Students' Understanding of Carbon Cycling using Topic Modeling. 2022-2023 International Travel Award, University of Iowa (\$1,200)
- You. H. (PI)**. Looking Beyond Disciplinary Silos: Development and Validation of an Interdisciplinary Science Assessment using an Item Response Model and Statistical Topic Model. Iowa Measurement Research Foundation (IMRF) (Not Selected)
- You. H.** Professional Development Award, UT-Austin (\$800)

2021

- You. H. (PI)**. Toward Interdisciplinary Learning: Assessing Interdisciplinary Understanding of College Students. Assessment Project Grant. Arkansas Tech University (03/04/20 - 03/05/21, \$4,916.70)
- You. H.** Texas New Scholar Award, UT-Austin (\$18,357)
- You. H.** Teresa Lozano Fellowship UT-Austin (\$1,300)

2020

You. H. (PI) & Patton, J. (Co-PI). Interdisciplinary Approach in Carbon Cycling: Experts' Conceptions and the US Standards. Undergraduate Research Grant, Arkansas Tech University (09/01/19 - 5/31/20, \$2,800)

You. H. Professional Development Grant, Arkansas Tech University (\$1,060)

2019

You. H. (PI). Revisiting The Relationship between Science Teaching Practice and Scientific Literacy from Global Perspective. AERA (Not Selected)

2018

You. H. (PI), Park, S. (Co-PI), & Kim, K. (Co-PI). What School-Level Factors Influence Mathematical and Scientific Literacy? A Multivariate Multi-Level Analysis using PISA 2015, AERA (Not Selected)

2015

You. H. (PI). Do Schools Make a Difference?: Exploring School Effects Mathematics Achievement in PISA 2012 using Hierarchical Linear Modeling. Korea Institute Curriculum And Evaluation (5/01/15 ~ 12/01/15, \$2,000)

You. H. The Continuing Fellowship, UT-Austin (\$16,000)

You. H. Graduate Student Research Award (Second Place), Korea Institute Curriculum and Evaluation

You. H. The Jewel Popham Raschke Endowed Presidential Scholarship in Mathematics Education, UT-Austin (\$5,000)

You. H. Graduate Student Professional Development Award, UT-Austin (\$445)

2014

You. H. The Joseph L. Henderson and Katherine D. Henderson Foundation Scholarship, UT-Austin (\$2,000)

You. H. The Education Annual Fund Endowed Presidential Scholarship, UT-Austin (\$2,500) **You. H.** Faculty/Student Collaborative Award, UT-Austin (\$899)

You. H. Graduate Student Professional Development Award, UT-Austin (\$250)

2013

You. H. Carolyn J. and John H. Young Endowed Presidential Fellowship in Education, UT-Austin (\$3,500)

2012

You. H. Faculty/Student Collaborative Award, UT-Austin (\$300)

You. H. Faculty/Student Collaborative Award, UT-Austin (\$728)

You. H. The Joseph L. Henderson and Katherine D. Henderson Foundation Scholarship, UT-Austin

(\$2,000)

2011

You. H. The Joseph L. Henderson and Katherine D. Henderson Foundation Scholarship, UT-Austin (\$1,000)

You. H. Joe R. & Teresa Lozano Long Graduate Fellowship Fund, UT-Austin (\$1,000)

2010

You. H. The Joseph L. Henderson and Katherine D. Henderson Foundation Scholarship, UT-Austin (\$1,000)

2003 ~ 2006 Out of the U.S.

Graduate School Scholarship, Yonsei University (\$2,400)

Sinyang Scholarship, Sinyang Cultural Foundation (\$2,500)

University Designated Scholarship, Yonsei University (\$6,000)

University Designated Scholarship, Yonsei University (\$1,500)

Students Union & Yonsei Cooperation Scholarship, Yonsei University (\$800)

University Designated Scholarship, Yonsei University (\$800)

■ TEACHING EXPERIENCE

K-12 Teaching Experience

Assistant Teacher, Research in Life Science Summer Camp, UT-Austin, 2015

Volunteer Math Teacher, Dobie Middle School, Austin, TX, 2014-2015

Science Teacher, Shin Myeong Public Middle School, Seoul, 2008-2010

Classes taught: Integrated Science, General Science, Biology, and Chemistry

Science Teacher, Dolma High School (public), Seoul, 2007-2008

Classes taught: General Science

Teacher Certificate

Missouri Provisional Teacher Certificate, General Science Grade 5-12 , 2018

Teacher Certificate of Secondary General Science, Ministry of Education, South Korea, 2008

College/University Teaching Experience

Assistant Professor, University of Iowa, 2022-present

EDTL 3165 Elementary Science Methods I, Fall 2022-Spring 2025

SIED 3164 I Intro Global Scientific Challenges (GE course), Spring 2025-

Assistant Professor, Arkansas Tech University, 2019-2022

PHSC 3243 Integrating the Three Dimensions of Science

PHSC 3253 Teaching Methods for STEM

CHEM 1111 Survey of Chemistry Laboratory

Teaching Assistant, University of Texas at Austin, 2012-2022

EDP 380C Fundamental Statistics: Spring 2022

EDP 380C Survey: Multivariate Method: Spring 2021

BIO 311C Introduction Biology I: Spring 2016/Summer 2015

BIO 311C Introduction Biology I in Texas Interdisciplinary Plan (TIP): Spring 2015

NTRI07L Introductory Food Science Laboratory: Fall 2014

NTR 366L Research Methods in Nutritional Science: Spring 2014/Fall 2013/Spring 2013

EDC 365C Knowing and Learning: Fall 2012

Teaching Survey Results (Overall Average): 4.85/5

■ PROFESSIONAL SERVICE

CO-Editor-in-Chief

Asia-Pacific Science Education (Scopus indexed, Published by SpringerOpen in partnership with *Korean Association for Science Education*), 2026~

Guest Editor

Education Sciences, 2025~

Journal Board Member

Journal of Research in Science Teaching (Top-Ranked Journal in Science Education), 2024-present

Association Committee

National Association for Research in Science Teaching Early Career Research Award Subcommittee, 2024-present

Korean Educational Learning Science Mentoring Committee, 2024-present

Korean Edutech/Learning Sciences Researcher Network Mentoring Committee, 2025-present

Journal/Conference Reviewer

Chemistry Education Research and Practice, 2023-present

Sage Open (article editor), 2023-present

School Science and Mathematics Reviewer, 2021-present

Journal of Research in Science Teaching Reviewer, 2018-present

Science Education Reviewer, 2018-present

Southeast Asian Journal of STEM Education Reviewer, 2020-present

Universal Journal of Educational Research Reviewer, 2020-present

Science and Education (SCED) Reviewer, 2018-present

International Journal of Science Education Reviewer, 2017-present

EURASIA Journal of Mathematics, Science and Technology Education Reviewer, 2018-present

NARST Conference Reviewer, 2011-present

AERA Conference Division C, D, and K Reviewer, 2011-present

International Conference of the Learning Sciences (ICLS) Conference Reviewer, 2012-present

Item Development Consultant

University of Wisconsin-Madison, Institute for Innovative Assessment, 2018

Educational Testing Service (ETS) - Learning Sciences/Research Division, 2017

Measurement Incorporated (MI) for the NGSS in K-12, 2016-2017

Statistics Consulting

Department of Mechanical and Aerospace Engineering, New York University, 2017-present

Department of Curriculum and Instruction, University of Texas at Austin, 2010-2016

College Student Assessment

Texas (CRAFT) grader, the School of Undergraduate Studies, UT-Austin, 2013 Project “College Readiness”

Professional Membership

National Science Teaching Association (NSTA), 2012-present

National Association for Research in Science Teaching (NARST), 2010-present

American Educational Research Association (AERA), 2010-present

International Conference of the Learning Sciences (ICLS), 2012-present

Society for the Advancement of Biology Education Research (SABER), 2017-present

■ ADMINISTRATIVE AND COMMITTEE SERVICE

University of Iowa

University

Non-Resident Classification Review committee 2023-present

First Generation Mentor 2023-present

Departmental

PDQ / NOC Ad Hoc Committee, 2023

Search Committee, Elementary Education Faculty, 2023

Search Committee, Math Education Faculty, 2025

MAT Student comprehensive committee 2023~

Doctoral Student Comprehensive Exam Committee, 2023-

Mandy Dunphy

Stephanie Erps

Alison Warren

Li Zhu

Zeynep Montesoglu

Mohammed Humaidi

Doctoral Student Dissertation Committee, 2023-

Li Zhu (co-chair)

Mohammed Humaidi (co-chair)

Dennis Kwaka (co-chair)

Zeynep Montesoglu (member)

Chris Like (member)

Development of a New CLAS (General Education) Course: “Intro to Global Socioscientific Challenges”

Proposal for a New Graduate Course: “Quantitative Reasoning in Educational Research”

■ COMMUNITY SERVICE

Book Fair Planning Committee Member, Cedar Hill elementary school, 2023-present

Volunteer Teacher, Kansas City Korean Language Institute, 2022-present

Referee, Arkansas Junior Academy of Science , 2021

Volunteer Mentor, Honors Science Graduate Program, Russellville High School, AR. 2019-2022

Volunteer AP Chemistry Tutor, Underrepresented Minority Students (TX, MO, KS), 2010-2022

Volunteer Math Teacher, Dobie Public Middle School, Austin, TX, 2014-2015 Assistant Life

Science Teacher, Middle School Students in UTeach Summer Camp, 2015

Volunteer Teacher for Older Adults in a Lifelong Science Program, Mapo Evening School, Seoul
2006-2007

Volunteer Organic Chemistry Tutor for Low Performers in College, Yonsei Center for Learning &
Teaching, 2004-2005

■ STATISTICAL SOFTWARE PROFICIENCY

Proficient with:

R – Data Management and Statistical Analyses
Python – Data Management and Statistical Analyses
SPSS – Statistical Analyses
Mplus – Structural Equation Modeling and/or Factor Analyses
HLM – Hierarchical Linear Modeling
WINSTEP – Rasch Modeling
FACETS – Rasch Modeling
IRTPRO – Item Response Theory

Experience with:

SQL – Data Management
SAS – Statistical Analyses
Tableau – Data Visualization
NVivo – Qualitative analyses
QDA Miner – Qualitative analyses

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